

Technology Terms Contract

- CURRENT STATE AND FUTURE STRATEGY
- Terms are grouped by platform layer and proof status
 - Current terms describe live or repo/admin verified behavior
 - Candidate terms describe gated implementation paths
 - Research-only terms describe Swiss RAvaFcast and Himalayan partner-evidence work
 - Future terms describe architecture direction, not completed product

Evidence level: Artifact/doc proof only

INTERPRETATION

CURRENT STATE

Evidence now

Use only hosted, admin, repo, and artifact proof that exists today.

GATED

Promotion rules

MTS-LSTM, SAR, TreeSHAP refresh, and runout physics require explicit evidence before promotion.

FUTURE STRATEGY

Validation program

Scientist review decides which technical paths become stronger operational claims.

Published Forecast Terms

CURRENT STATE AND FUTURE STRATEGY

- `Forecast Run`: packaged publication event with metadata and freshness
- `Forecast Grid`: spatial cells carrying danger, uncertainty, explanation, and optional runout references
- `Artifact Manifest`: small index describing larger forecast payloads
- `Batch-First Delivery`: heavy work runs upstream before the user opens the website

Evidence level: `Hosted production plus Repo/admin verified`

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Public Platform Terms

CURRENT STATE AND FUTURE STRATEGY

- **React**: reusable interface components
- **Vite**: production frontend build tool
- **Leaflet**: map interaction layer
- **Recharts**: dashboard and admin charting layer
- **Single-Page Application**: routed app experience without full page reloads

Evidence level: Repo/admin verified plus Hosted production

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Supabase And Access- Control Terms

CURRENT STATE AND FUTURE STRATEGY

- Supabase: Postgres, Auth, Storage, and Edge Functions
- Postgres: structured records for runs, reports, jobs, and model status
- Row-Level Security: database access boundaries
- Edge Function: lightweight server-side task surface
- Admin Route: restricted observability and review lane

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Offline And Field Workflow Terms

CURRENT STATE AND FUTURE STRATEGY

- PWA: installable/offline-capable web pattern
- Service Worker: browser background process for offline behavior
- IndexedDB: local structured browser storage
- Background Sync: deferred replay pattern when connectivity returns
- Future strategy: field-device smoke tests and operator review workflow

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Avalanche Communication Terms

CURRENT STATE AND FUTURE STRATEGY

- `Forecast Bulletin`: structured danger, problem, uncertainty, and terrain summary
- `EAWS-style experimental`: familiar structure without authority-grade equivalence
- `APT`: avalanche-prone terrain used to avoid false low-risk semantics
- `Masked Terrain`: out-of-scope terrain displayed separately from low danger
- `Uncertainty Cue`: visible signal that confidence is limited

Evidence level: `Hosted production` plus `Artifact/doc proof` only

INTERPRETATION

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Active Model Terms

CURRENT STATE AND FUTURE STRATEGY

- **Random Forest**: current active baseline model family
- **Calibration**: making predicted probabilities match observed frequency
- **Brier Score**: probability calibration metric
- **PSS**: rare-event discrimination metric
- **KMeansSMOTE** and **Recursive Feature Elimination**: training and feature-discipline concepts
- **RF4 reproduction**: research-only Swiss danger-level signal, not paper parity and not Himalayan proof

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Explainability Terms

CURRENT STATE AND FUTURE STRATEGY

- SHAP: family of feature-attribution methods
- TreeSHAP: SHAP method specialized for tree models
- Current active artifact: heuristic_fallback
- Current status: implemented path, not stronger active-run proof
- Future strategy: active artifact refresh with computed or cached TreeSHAP values

Evidence level: Repo/admin verified plus Artifact/doc proof only

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Candidate Sequence-Model Terms

CURRENT STATE AND FUTURE STRATEGY

- LSTM: sequence neural network for temporal patterns
- MTS-LSTM: multi-time-scale sequence model candidate
- Current status: repo/admin verified candidate
- Promotion rule: benchmark, stability, and release gates must pass first
- Future strategy: shadow runs before any public scorer change

Evidence level: Repo/admin verified plus Artifact/doc proof only

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SAR And Remote-Sensing Terms

CURRENT STATE AND FUTURE STRATEGY

- SAR: Synthetic Aperture Radar, weather-resistant satellite imaging
- Sentinel-1: SAR mission context
- U-Net: segmentation model family for masks and pixel regions
- Current status: shadow/candidate workflows and coverage caveats
- Current boundary: shadow-gated, no SAR production claim
- Future strategy: label coverage, revisit handling, shadow/layover review, and qualification artifacts

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Modal.com And GPU Terms

CURRENT STATE AND FUTURE STRATEGY

- Modal.com: remote worker and artifact-volume platform
- GPU: parallel hardware used for heavy training or segmentation workloads
- Current Modal use: SAR training/segmentation and MTS-LSTM training candidates
- Current public scorer: Random Forest baseline outside Modal.com
- Future strategy: GPU-backed batch inference only if cost and performance evidence justify it

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Terrain, Runout, And Geospatial Terms

CURRENT STATE AND FUTURE STRATEGY

- DEM: digital elevation model for slope, elevation, and aspect
- Alpha-Beta Runout: terrain-based runout estimation method
- WhiteboxTools: geospatial toolkit for future runout validation
- PostGIS: proposed geospatial database extension
- Current state: analytical Alpha-Beta fallback; Whitebox smoke proof separate from active public runout proof

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Data Lineage And Synthetic Boundary Terms

● TECHNICAL EVIDENCE

● RESEARCH AGENDA

CURRENT STATE AND FUTURE STRATEGY

- Data Lineage: records where a training or publication input came from
- source_ref: SHA-256-qualified pointer from each partner evidence row to a reviewed source package
- partner_source_manifest: partner-declared source owner, license, review, date range, and evidence-package reference
- triage_artifact_manifest: generated inventory of output files, sizes, hashes, and purposes
- Current active proof: non-synthetic full-grid technical publication; synthetic support remains visibly flagged where present
- Future strategy: strict separation of technical proof, partner intake, training eligibility, and customer-visible validation data
- Cleanup rule: isolate synthetic seeds, weights, manifests, and claims behind explicit lineage fields

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Standards And Interoperability Terms

CURRENT STATE AND FUTURE STRATEGY

- COG: Cloud Optimized GeoTIFF for efficient future raster access
- OGC APIs: future geospatial interoperability direction
- CAP: Common Alerting Protocol for future alert integration discussions
- D_forecast: raw or official forecast label that may contain human forecast noise
- D_tidy: quality-controlled danger label backed by nowcast, observer, event, or reanalysis evidence
- GPxyz: Gaussian-process interpolation using latitude, longitude, and elevation in the Swiss RAvaFcast setting
- Refined discretization: expected-danger thresholding learned only from training or out-of-bag distributions
- Current state: not a statutory warning-service or standards-certified alerting system
- Future strategy: standards mapping after scientist and operator validation

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Proof Buckets And Release Gates

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- `Hosted production`: visible on the deployed website
- `Repo/admin verified`: inspectable in source, artifacts, tests, or admin
- `Artifact/doc proof only`: documented or proposed, not public proof
- `Candidate/gated`: implemented path blocked from promotion until evidence passes
- `Research-only`: Swiss RAVaFcast and Himalayan partner intake artifacts that must keep `production_scoring_allowed=false`
- `Claims blocked`: `himalayan_accuracy_claim_allowed=false` until local evidence, holdout, scientist review, and release gates pass
- Future strategy: move terms upward only when artifacts and scientist review support promotion

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